

AWS Skill Builder

Exam Review Guide

AWS Certified Machine Learning — Specialty

65 Questions with All Options, Correct Answers & Explanations

Black & White Edition | One Question Per Page

For Offline Review & Study

Question 1

A company has trained an ML model in Amazon SageMaker AI. The company needs to host the model to provide inferences in a production environment. The model must be highly available and must respond with minimum latency. The size of each request will be between 1KB and 3MB. The model will receive unpredictable bursts of requests during the day. The inferences must adapt proportionally to the changes in traffic.

✓ **A. Create a SageMaker AI real-time inference endpoint. Configure auto scaling. Configure the endpoint to present the existing model.**

B. Use Spot Instances with a Spot Fleet behind an Application Load Balancer (ALB) for inferences. Use the ALBRequestCountPerTarget metric as the metric for auto scaling.

C. Install SageMaker AI Operators on an Amazon Elastic Kubernetes Service (Amazon EKS) cluster. Deploy the model in Amazon EKS. Set the Horizontal Pod Autoscaler to scale replicas based on the memory metric.

D. Deploy the model on an Amazon Elastic Container Service (Amazon ECS) cluster. Use ECS scheduled scaling that is based on the CPU of the ECS service.

Correct Answer: A

Explanation:

Real-time endpoints with auto scaling provide hosting capabilities for ML models in production. Additionally, real-time endpoints with auto scaling provide low-latency responses for real-time inference requests. Real-time inference can efficiently handle request sizes between 1KB and 3MB. Auto scaling ensures that the endpoint adapts to unpredictable bursts of traffic during the day by adjusting capacity based on demand. This solution provides built-in integration with SageMaker AI trained models. This solution provides an ML-optimized environment.

Question 2

A company has deployed an ML model by using an Amazon SageMaker AI endpoint. The company needs to deploy the latest version of the model in several steps over several days. The company will start by exposing 10% of users to the new model. Each day, the company will increase user exposure to the new model by 10% until 100% of users are using the new model. If the new version encounters errors, the company needs the capability to roll back immediately to the previous version.

A. Use a rolling deployment.

✓ **B. Use a blue/green deployment with linear traffic shifting.**

C. Use a blue/green deployment with canary traffic shifting.

D. Use a blue/green deployment with all-at-once traffic shifting.

Correct Answer: B

Explanation:

You can use a blue/green deployment with linear traffic shifting to gradually shift traffic from an existing environment (blue) to a new environment (green) over time. This method meets the requirement of starting with 10% user exposure and increasing 10% each day until you reach 100%. Additionally, you can use this method to roll back to the previous environment (blue) if issues occur.

Question 3

A restaurant company is building an ML model by using Amazon SageMaker AI to predict daily sales. The sales system generates data in CSV format. The company has 6,000 restaurants and generates many files each day. The company performs queries on the data. An ML engineer must build a solution to prepare, move, and integrate the data to use in the ML model. The ML engineer wants to use a serverless service and optimize query performance.

- A. Store the data in Amazon Elastic File System (Amazon EFS). Use AWS Glue to extract, transform, and load (ETL) the data in JSON format.
- B. Store the data in Amazon Elastic Block Store (Amazon EBS). Use SageMaker Ground Truth to extract, transform, and load (ETL) the data in CSV format.
- C. Store the data in Amazon S3 in CSV format. Use SageMaker Model Monitor to define and maintain a schema for the data.

✓ **D. Store the data in Amazon S3 in CSV format. Use AWS Glue to extract, transform, and load (ETL) the data in Apache Parquet format.**

Correct Answer: D

Explanation:

Amazon S3 is an object storage service. AWS Glue is a serverless data integration service that you can use for ETL jobs. You can use AWS Glue to convert CSV files into Parquet format. Parquet is an open source columnar data format that is suitable to save file space and optimize query performance.

Question 4

A company has an ML model that uses historical transaction data to predict customer behavior. An ML engineer is optimizing the model in Amazon SageMaker AI to enhance the model's predictive accuracy. The ML engineer must examine the input data and the resulting predictions to identify trends that could skew the model's performance across different demographics.

- A. Create AWS Lambda functions to automate data preprocessing and to ensure consistent quality of input data for the model.
- B. Use Amazon CloudWatch to monitor network metrics and CPU metrics for resource optimization during model training.
- C. Create AWS Glue DataBrew recipes to correct the data based on statistics from the model training.
- ✓ **D. Use SageMaker Clarify to evaluate the model and training data for underlying patterns that might affect accuracy.**

Correct Answer: D

Explanation:

You can use SageMaker Clarify to understand and explain how ML models make predictions. SageMaker Clarify provides tools to detect bias in training data and model predictions, evaluate feature importance, and generate model explainability reports. SageMaker Clarify can analyze the input data and resulting predictions to identify trends that could skew the model's performance across different demographics.

Question 5

A company has an Amazon S3 bucket that contains 1TB of files from different sources. The S3 bucket contains the following file types in the same S3 folder: CSV, JSON, XLSX, and Apache Parquet. An ML engineer must implement a solution that uses AWS Glue DataBrew to process the data. The ML engineer also must store the final output in Amazon S3 so that AWS Glue can consume the output in the future.

- A. Use DataBrew to process the existing S3 folder. Store the output in YAML format.
- B. Separate the data into a different folder for each file type. Use DataBrew to process each folder individually. Store the output in YAML format.
- ✓ **C. Separate the data into a different folder for each file type. Use DataBrew to process each folder individually. Store the output in Apache Parquet format.**
- D. Use DataBrew to process the existing S3 folder. Store the output in Apache Parquet format.

Correct Answer: C

Explanation:

DataBrew is a visual data preparation tool that you can use to clean and normalize data without writing code. To process data in DataBrew, you must organize the file types of different formats into separate folders by type. Apache Parquet is a columnar storage format that provides compression and query performance. DataBrew can process each file type correctly. This solution ensures that the output is optimized for future consumption by AWS Glue.

Question 6

An ML engineer is using Amazon SageMaker AI to build and compare binary classification models for a company. The company needs a model that returns results with the fewest possible false positives.

- A. Use DeepAR forecasting as the algorithm. Use recall as the metric.
- ✓ **B. Use LightGBM as the algorithm. Use precision as the metric.**
- C. Use CatBoost as the algorithm. Use recall as the metric.
- D. Use Random Cut Forest (RCF) as the algorithm. Use precision as the metric.

Correct Answer: B

Explanation:

LightGBM is a gradient boosting framework that you can use for binary classification tasks. Precision measures the proportion of true positive predictions among all positive predictions. Therefore, precision is the appropriate metric to minimize false positives.

Question 7

A company builds and deploys ML models by using Amazon SageMaker AI. The models are used across multiple departments. The company allocates a monthly budget that each department can spend on AWS services. The company wants to ensure that each department knows when the department's monthly spend is likely to exceed the allocated budget.

- A. Create a budget in AWS Budgets with an auto-adjusting method. Specify a filter based on AWS generated cost allocation tags that are associated with each department. Configure the budget to trigger an alert and notify the department based on the actual monthly spend.
- B. Create a budget in AWS Budgets with an auto-adjusting method. Specify a filter based on AWS generated cost allocation tags that are associated with each department. Create budget reports for each department and configure AWS Budgets to deliver daily reports.
- C. Create a cost report for each department by using AWS Cost Explorer. Filter the cost report based on user-defined cost allocation tags that are associated with each department. Grant each department access to its own Cost Explorer report to review and predict when each department will exceed its own budget.

✓ **D. Create a budget in AWS Budgets with a fixed method. Specify a filter based on user-defined cost allocation tags that are associated with each department. Configure the budget to trigger an alert and notify the department based on the forecasted monthly spend.**

Correct Answer: D

Explanation:

You can use AWS Budgets to track and take actions based on AWS spend. You can use a fixed method for AWS Budgets to define a static budget every month. You can use user-defined cost allocation tags to monitor costs based on tags that you create, such as department ID or project ID. You can use AWS Budgets to establish filters based on tags and to trigger alerts. By configuring an alert based on the forecasted monthly spend, you can anticipate the likelihood of your monthly spend exceeding a defined threshold.

Question 8

A company wants to improve its fraud detection capabilities. The company must implement a highly available streaming solution on AWS to ingest credit card data in real time to detect fraudulent transactions.

A. Use Amazon Kinesis Data Streams to ingest the data. Configure Amazon Bedrock to read from the Kinesis data stream to detect fraudulent transactions.

✓ **B. Use Amazon Kinesis Data Streams to ingest the data. Use Amazon Managed Service for Apache Flink to analyze data. Use Amazon SageMaker AI to detect fraudulent transactions.**

C. Use Amazon SageMaker Data Wrangler to ingest the data in real time. Configure SageMaker Autopilot to perform the analysis to detect fraudulent transactions.

D. Use Amazon SageMaker Autopilot to automatically ingest the data, build a model, and analyze the data to detect fraudulent transactions.

Correct Answer: B

Explanation:

Kinesis Data Streams can ingest credit card data in real time. Then, Managed Service for Apache Flink can analyze the streaming data. Finally, you can use SageMaker AI to detect fraudulent transactions by using ML models. This solution provides a highly available streaming architecture for real-time fraud detection.

Question 9

A company runs training jobs on Amazon SageMaker AI by using a compute optimized instance. Demand for training runs will remain constant for the next 55 weeks. The instance needs to run for 35 hours each week. The company needs to reduce its model training costs.

A. Use a SageMaker training plan to reserve the compute optimized instance for training. Specify the instance type. Run the training job.

B. Use the heterogeneous cluster feature of SageMaker Training. Configure the `instance_type`, `instance_count`, and `instance_groups` arguments to run training jobs.

✓ **C. Opt in to a SageMaker AI Savings Plan with a 1-year term and an All Upfront payment. Run a SageMaker Training job on the instance.**

D. Use a serverless endpoint with a provisioned concurrency of 35 hours for each week. Run the training on the serverless endpoint.

Correct Answer: C

Explanation:

SageMaker AI Savings Plans provide a flexible pricing model with savings on SageMaker AI usage. With Savings Plans, you must commit to a consistent amount of usage for a 1-year or 3-year term. In this scenario, the company has a consistent, predictable usage pattern of 35 hours each week for 55 weeks. You can opt for a Savings Plan with a 1-year term and an All Upfront payment to receive a discount on SageMaker AI training costs.

Question 10

A company deployed an ML model into production. A data team needs to periodically ingest and transform updated datasets to support batch inference. The data team creates an Amazon SageMaker Data Wrangler data flow to transform the raw data into features. Select and order the correct steps from the following list to make the data available for batch inferencing. Each step should be selected one time or not at all.

- ✓ **1. Specify a feature group as a data flow destination.**
- ✓ **2. Run a SageMaker Data Wrangler processing job.**
- ✓ **3. Read features from the offline SageMaker Feature Store.**

Correct Answer: 1 → 2 → 3

Explanation:

Correct order: 1. Specify a feature group as a data flow destination. 2. Run a SageMaker Data Wrangler processing job. 3. Read features from the offline SageMaker Feature Store.

Question 11

A company is deploying a computer vision model in retail stores. The model is for live self-checkout fraud detection and is deployed on an Amazon SageMaker AI real-time inference endpoint. After updates to the model, the company must validate the updated model on 10% of customers before making the model immediately available to all customers. The company has already created Amazon CloudWatch alarms to monitor the endpoint. The company will use the SageMaker UpdateEndpoint API operation for the deployment.

✓ **A. Use a canary deployment. In the CanarySize field, specify CAPACITY_PERCENT with a value of 10.**

B. Use a linear deployment. In the LinearStepSize field, specify CAPACITY_PERCENT with a value of 10.

C. Use a canary deployment. In the CanarySize field, specify CAPACITY_PERCENT with a value of 90.

D. Use a linear deployment. In the LinearStepSize field, specify CAPACITY_PERCENT with a value of 90.

Correct Answer: A

Explanation:

In canary deployment, you deploy the new model version to a small group of customers before you roll out the new model version to all customers. A canary deployment with the CanarySize field set to CAPACITY_PERCENT with a value of 10 meets the requirements. This configuration will validate the updated model on 10% of all incoming requests before full rollout. This configuration directs 10% of traffic to the new model version. This configuration keeps 90% of the traffic on the current production version. Therefore, this configuration provides the opportunity for controlled validation.

Question 12

A binary classification model's performance has decreased significantly during the past few days. An ML engineer needs to adjust the hyperparameters of the model. The underlying model type must remain the same.

- A. SageMaker AI notebook with a custom random search.
- B. SageMaker Autopilot with the default configuration.
- C. SageMaker Autopilot with the TabularJobConfig ProblemType attribute set to BinaryClassification.

✓ **D. SageMaker AI automatic model tuning.**

Correct Answer: D

Explanation:

You can use SageMaker AI automatic model tuning to adjust hyperparameters with minimal effort. You can use SageMaker AI automatic tuning to optimize hyperparameters while keeping the underlying model type the same. This functionality automates the process to identify the best hyperparameter configuration. Therefore, this solution significantly reduces effort compared to manual approaches.

Question 13

A company needs to use an Amazon SageMaker AI built-in algorithm for supervised training to predict whether a customer will buy a product.

- A. Random Cut Forest (RCF)
- B. K-means
- ✓ C. K-nearest neighbors (k-NN)
- D. Support Vector Regression (SVR)

Correct Answer: C

Explanation:

The k-NN algorithm is a supervised learning algorithm that you can use for classification tasks. An example of a classification task is predicting whether a customer will buy a product. The k-NN algorithm meets the requirement for supervised training and can make predictions based on labeled historical data. The k-NN algorithm can find the closest data points to a new input and can use labels to make a prediction. Therefore, this solution is appropriate for binary classification problems like customer purchase prediction.

Question 14

An ML engineer notices class imbalance in an image classification training dataset.

- A. Transform some of the images in the dataset.
- B. Apply random data splitting on the dataset.
- C. Reduce the size of the dataset.
- ✓ **D. Apply random oversampling on the dataset.**

Correct Answer: D

Explanation:

Class imbalance occurs when one class in a dataset significantly outweighs the other class. You can apply random oversampling on the dataset to effectively address class imbalance in image classification. This technique involves creating additional samples of the minority class. For this technique, you can duplicate existing samples or generate synthetic samples. By increasing the representation of the underrepresented class, random oversampling helps balance the class distribution. This approach directly addresses the class imbalance problem without altering the original data.

Question 15

A manufacturing company uses an ML model to determine whether products meet a standard for quality. The model produces an output of "Passed" or "Failed". Robots separate the products into the two categories by using the model to analyze photos on the assembly line.

- A. Perplexity
- B. Bilingual Evaluation Understudy (BLEU) score
- C. Root mean squared error (RMSE) and mean absolute percentage error (MAPE)
- D. Precision and recall
- E. Accuracy and F1 score

Correct Answer: D, E

Explanation:

This problem is a binary classification task (pass/fail). Precision measures the proportion of products that are classified as "Passed" that actually meet the standard for quality. Recall measures the proportion of products with the label "Passed" that are correctly classified as "Passed". Accuracy measures the overall proportion of products that are correctly classified as "Passed" or "Failed". The F1 score is the harmonic mean of precision and recall. The F1 score provides a single metric that balances both measures. These metrics are useful for evaluating the overall performance of the model in correctly classifying products.

Question 16

A company wants to develop an ML model by using tabular data from its customers. The data contains meaningful ordered features with sensitive information that should not be discarded. An ML engineer must ensure that the sensitive data is masked before another team starts to build the model.

- A. Run an Amazon EMR job to change the sensitive data to random values.
- B. Use Amazon Macie to categorize the sensitive data.
- ✓ **C. Prepare the data by using AWS Glue DataBrew.**
- D. Run an AWS Batch job to change the sensitive data to random values.

Correct Answer: C

Explanation:

DataBrew is a data preparation service that includes built-in transformations. You can use a built-in transformation to mask data while preserving the statistical properties and relationships in the data.

Question 17

A financial services company is migrating its data science environment to an Amazon SageMaker Studio domain. The data necessary for exploratory analysis is already stored in an Amazon S3 bucket. All network traffic must stay in the company's private network. The company must configure the SageMaker Studio domain and networking appropriately.

- A. Create the SageMaker Studio domain in VPC only mode.
- B. Use a public subnet to run SageMaker Studio and to access the data in Amazon S3.
- C. Use a NAT gateway to run SageMaker Studio and to access the data in Amazon S3.
- D. Create the SageMaker Studio domain with the `EnableNetworkIsolation` parameter set to `True`.
- E. Use interface VPC endpoints to run SageMaker Studio and to access the data in Amazon S3.

Correct Answer: A, E

Explanation:

Creating a SageMaker Studio domain in VPC only mode prevents internet access to SageMaker Studio notebooks. In this case, SageMaker Studio communicates with other hosts and AWS services through a customer VPC and not through the internet. Interface VPC endpoints allow SageMaker Studio to access other AWS services from a customer VPC without requiring internet access. You can use interface VPC endpoints to allow SageMaker Studio to access Amazon S3, the SageMaker runtime, and the SageMaker API.

Question 18

A company is using Amazon SageMaker AI to develop an ML model. The model uses tabular data and image data. An ML engineer needs to implement a solution to prepare and visualize the data for the model.

A. Use SageMaker Processing to prepare and visualize the data.

✓ **B. Use SageMaker Data Wrangler to prepare and visualize the data.**

C. Use SageMaker batch transform to prepare the data. Use SageMaker Data Wrangler to visualize the data.

D. Use SageMaker Data Wrangler to prepare the data. Use Amazon QuickSight to visualize the data.

Correct Answer: B

Explanation:

You can use SageMaker Data Wrangler to import, prepare, transform, and analyze data in a low-code no-code interface. SageMaker Data Wrangler provides a unified solution for data preparation and visualization. SageMaker Data Wrangler provides built-in support for tabular and image data.

Question 19

A bank has used Amazon SageMaker AI to develop a credit risk scoring model to evaluate loan applications. The model is deployed on a SageMaker AI real-time endpoint with the data capture feature enabled. The bank needs real-time explanations about the model's predictions.

✓ **A. Update the SageMaker AI endpoint with an endpoint configuration that includes the SageMaker Clarify ExplainerConfig option. Extract the model predictions and Shapley Additive Explanations (SHAP) feature attributions from the data that is returned from the endpoint.**

B. Create a SageMaker Jupyter notebook. Import the Shapley Additive Explanations (SHAP) library. Write Python code to process each captured input data point and to generate the SHAP feature attributions for each model prediction.

C. Run a SageMaker Clarify processing job on the endpoint inputs. Configure the processing job to return model explainability. Access the explainability report. Extract the general Shapley Additive Explanations (SHAP) feature attributions of the model.

D. Run a SageMaker Clarify processing job on the endpoint inputs. Configure the processing job to return model explainability and the local Shapley Additive Explanations (SHAP) values of each prediction. Access the report outputs in Amazon S3. Extract the SHAP feature attributions for each model prediction.

Correct Answer: A

Explanation:

You can update the SageMaker AI endpoint with the SageMaker Clarify ExplainerConfig option. Then, you can obtain both model predictions and SHAP feature attributions in real time from the endpoint. This solution meets the real-time requirement to evaluate loan applications and provides immediate explanations for each prediction.

Question 20

A company uses AWS provided libraries during Amazon SageMaker AI training jobs. The SageMaker AI domain is configured to use public subnets. The company does not want SageMaker AI to collect metadata about the use of the AWS provided libraries.

A. Set the subnet used by SageMaker AI to be private by using the VpcConfig option in the CreateTrainingJob API.

✓ **B. Set the OPT_OUT_TRACKING environment variable to 1 in the SageMaker AI CreateTrainingJob API.**

C. Disable the use of a VPC from SageMaker AI by using the VpcConfig option in the CreateTrainingJob API.

D. Set the OPT_OUT_TRACKING environment variable to 0 in the SageMaker AI CreateTrainingJob API.

Correct Answer: B

Explanation:

You can set the OPT_OUT_TRACKING environment variable to 1 in the SageMaker AI CreateTrainingJob API to prevent SageMaker AI from collecting metadata about the use of AWS provided libraries. This directly addresses the company's requirement to avoid collecting metadata during training jobs. You can use this environment variable to opt out of SageMaker AI metadata collection for AWS provided libraries.

Question 21

A company is using an AWS Lambda function to monitor the metrics from an ML model. An ML engineer needs to implement a solution to send an email message when the metrics breach a threshold.

- A. Log the metrics from the Lambda function to AWS CloudTrail. Configure a CloudTrail trail to send the email message.
- B. Log the metrics from the Lambda function to Amazon CloudFront. Configure an Amazon CloudWatch alarm to send the email message.
- ✓ **C. Log the metrics from the Lambda function to Amazon CloudWatch. Configure a CloudWatch alarm to send the email message.**
- D. Log the metrics from the Lambda function to Amazon CloudWatch. Configure an Amazon CloudFront rule to send the email message.

Correct Answer: C

Explanation:

CloudWatch is a monitoring and observability service that monitors AWS resources and applications. CloudWatch provides data and actionable insights to monitor applications, to respond to system-wide performance changes, and to optimize resource utilization. Additionally, CloudWatch can provide a unified view of operational health. You can use CloudWatch to collect metrics, to set alarms, and to invoke automated actions such as sending email notifications. Lambda can write to CloudWatch. CloudWatch alarms can alert and send emails when a metrics threshold is breached.

Question 22

A company receives hundreds of resumes every day. The company stores the resumes in an Amazon S3 bucket. The company needs to deploy an automated and scalable solution to detect and extract personally identifiable information (PII) from PDF resume files. The solution must also detect and provide PII entities before saving the result back to the S3 bucket.

- A. Create an Amazon Transcribe batch translation job. Start an asynchronous job to detect the PII entities.
- B. Call the Amazon Textract asynchronous API operations to detect the PII entities.
- ✓ **C. Create an Amazon Comprehend analysis job. Start an asynchronous job to detect the PII entities.**
- D. Create an Amazon Personalize batch inference job. Start an asynchronous job to detect the PII entities.

Correct Answer: C

Explanation:

Amazon Comprehend is a natural language processing (NLP) service that you can use to extract insights from documents. Amazon Comprehend can detect and redact PII in PDF files. You can use an Amazon Comprehend analysis job to parse multiple documents asynchronously. Then, you can save the results back to the S3 bucket.

Question 23

A bank is preparing data to train an ML model on Amazon SageMaker AI. The ML model will be used for loan approvals. The bank notices that the historical loan data for training contains many types of biases. An ML engineer must identify the features that contain bias with minimal operational overhead. Select and order the correct steps from the following list to identify features that contain bias with the LEAST operational overhead. Each step should be selected one time or not at all.

- ✓ **1. Perform an exploratory data analysis (EDA) by using SageMaker Data Wrangler to identify the features to analyze.**
- ✓ **2. Create and run a bias report by using SageMaker Data Wrangler and specify the features that were identified.**
- ✓ **3. Review the report by using SageMaker Data Wrangler.**

Correct Answer: 1 → 2 → 3

Explanation:

Correct order: 1. Perform an exploratory data analysis (EDA) by using SageMaker Data Wrangler to identify the features to analyze. 2. Create and run a bias report by using SageMaker Data Wrangler and specify the features that were identified. 3. Review the report by using SageMaker Data Wrangler.

Question 24

A company deployed an ML model in Amazon SageMaker AI over 6 months ago. The model provides product recommendations. The model contributed to increased sales in the first quarter. In the most recent quarter, the number of customers purchasing the model's recommended products has been steadily declining.

- A. Re-train the model periodically with only the original data. Apply regularization to improve prediction accuracy.
- B. Tune the hyperparameters periodically to reduce model complexity.
- ✓ **C. Re-train the model periodically with new and original data when a significant change in customer data is detected.**
- D. Train a new model with only new data. Use a different ML algorithm to handle model drift.

Correct Answer: C

Explanation:

For a model to effectively make accurate predictions, you must re-train the model when you detect drift. Re-training the model when you detect significant changes in customer data can help the model adapt to new patterns and trends in customer behavior. You should use both new and existing data. You should re-train with new data because the new data contains patterns that were not present in the original data. Additionally, you should re-train with existing data because you do not want to lose patterns that were present in the original data. This solution can mitigate the impact of model drift and improve the model's performance over time.

Question 25

A company needs to develop an ML model. The model must identify an item in an image and must provide the location of the item.

- A. XGBoost
- B. K-nearest neighbors (k-NN)
- ✓ **C. Object detection**
- D. Image classification

Correct Answer: C

Explanation:

Object detection identifies items in an image and provides the locations of the items. Object detection often provides the locations in the form of bounding boxes. You can use the object detection algorithm in SageMaker AI to identify an item in an image and provide the location of the item.

Question 26

A company plans to deploy an ML model for production inference on an Amazon SageMaker AI endpoint. The average inference payload size will vary from 100MB to 300MB. Inference requests must be processed in 60 minutes or less, and the endpoint must remain available.

✓ **A. Asynchronous inference**

B. Batch transform

C. Real-time inference

D. Serverless inference

Correct Answer: A

Explanation:

Asynchronous inference supports payloads up to 1GB. Therefore, this solution can accommodate the 100MB to 300MB payload size requirement. Asynchronous inference also supports long-running processes of up to 1 hour. Therefore, this solution meets the 60-minute processing time requirement. Asynchronous inference is suitable when you do not need immediate responses and you need to handle larger payloads with longer processing times.

Question 27

A company has a medical image diagnostics application that analyzes images that are up to 1MB in size. Hospitals use REST API calls to AWS to transmit images to CPU-based inference models that are hosted on Amazon SageMaker AI. The models take less than 1 minute to process each image. The models must respond with interpretations for the images in real time. The traffic volume to the models is unpredictable during working hours and is almost nonexistent after working hours.

✓ **A. Set up SageMaker Serverless Inference. Scale the underlying infrastructure based on workload requests.**

B. Set up SageMaker Asynchronous Inference. Queue the inference requests and respond within 1 hour.

C. Set up SageMaker AI batch transform to preprocess the images. Perform inference at the end of each day.

D. Host the models in a Docker container. Use SageMaker AI to perform real-time inference.

Correct Answer: A

Explanation:

SageMaker Serverless Inference provides real-time inference. Therefore, this solution meets the requirement for immediate responses to REST API calls. The serverless architecture automatically scales based on workload. Therefore, serverless inference can handle the unpredictable traffic during working hours efficiently. Serverless Inference is cost-effective because you pay for only the compute time that you use to process requests. Therefore, this solution is suitable for the scenario where traffic is almost nonexistent after working hours. Serverless inference is suitable for the 1MB image size because serverless inference is optimized for payloads under 4MB.

Question 28

A company needs to train an ML model that will use historical transaction data to predict customer behavior. Select the correct AWS service from the following list to perform each task on the data. Each service should be selected one time or not at all. (Note: The matching answer table was interactive and was not captured in the saved page.)

(Matching question - interactive table not captured in the report)

Correct Answer: N/A

Explanation:

Matching question - answer table was not captured in the Skill Builder report.

Question 29

A company wants a cost-effective way to train an ML model from AWS Marketplace algorithms by using Amazon SageMaker AI. The model is trained once each month and takes several hours. Training time is not a concern for the company.

✓ **A. Use SageMaker AI managed spot training.**

B. Use an Amazon EC2 Instance Savings Plan.

C. Use a SageMaker AI Savings Plan.

D. Use the Amazon EC2 On-Demand Instance pricing.

Correct Answer: A

Explanation:

You can use SageMaker AI to build, train, and deploy ML models. You can use managed EC2 Spot Instances to train the model. Managed spot training can optimize the cost of training models compared to On-Demand Instances. SageMaker AI can also manage spot interruptions for you.

Question 30

A company needs to ingest data from an Amazon Redshift cluster and a Snowflake data source into Amazon SageMaker Data Wrangler. The company also must transform the data to prepare the data for ML model training.

- A. Use AWS Glue to crawl the Redshift cluster and the Snowflake data source. Import the resulting data catalog into SageMaker Data Wrangler. Use the data preparation capabilities in SageMaker Data Wrangler to transform the data.
- B. Use Amazon Athena to directly query the Redshift cluster and the Snowflake data source. Use Athena also to transform the data. Import the query results into SageMaker Data Wrangler.
- ✓ **C. Connect to the Redshift cluster and the Snowflake data source by using separate data connections in SageMaker Data Wrangler. Import the data by using SQL queries. Use the data preparation capabilities in SageMaker Data Wrangler to transform the data.**
- D. Use Amazon Kinesis Data Streams to stream the data directly from the Redshift cluster and the Snowflake data source into an Amazon S3 bucket. Import the data from the S3 bucket into SageMaker Data Wrangler. Use the data preparation capabilities in SageMaker Data Wrangler to transform the data.

Correct Answer: C

Explanation:

SageMaker Data Wrangler supports direct connections to both Amazon Redshift and Snowflake through built-in connectors. You can write SQL queries to import data from the Amazon Redshift and Snowflake sources. Then, you can use the transformation capabilities of SageMaker Data Wrangler to prepare the data for ML model training.

Question 31

A company that has hundreds of data scientists is using Amazon SageMaker AI to create ML models. The models are in model groups in the SageMaker Model Registry. The data scientists are grouped into three categories: computer vision, natural language processing (NLP), and speech recognition. An ML engineer needs to implement a solution to organize the existing models into these categories to improve model discoverability at scale. The solution must not affect the integrity of the model artifacts and their existing structure.

A. Use SageMaker ML Lineage Tracking to automatically identify and tag which model groups should contain the models.

✓ **B. Create a SageMaker Model Registry collection for each of the three categories. Move the existing model groups into the collections.**

C. Create a custom tag for each of the three categories. Add the tags to the model packages in the SageMaker Model Registry.

D. Create a model group for each category. Move the existing models into these category model groups.

Correct Answer: B

Explanation:

SageMaker Model Registry collections are specifically designed to organize model groups into logical categories without affecting their existing structure. Collections will preserve the existing model group relationships. Additionally, collections can improve model discoverability at scale.

Question 32

An ML engineer needs to create a training environment for thousands of training jobs. Each training job runs for a short duration of time and uses the same configuration, such as instance type. Each job needs to start running as soon as the previous job finishes running. On some days, no training jobs run.

A. Create an Amazon Elastic Kubernetes Service (Amazon EKS) cluster that uses AWS Fargate. Use Apache Airflow to queue the training jobs.

✓ **B. Create Amazon SageMaker AI training jobs. Set the `keep_alive_period_in_seconds` value for each job.**

C. Create an Amazon SageMaker HyperPod cluster. Queue the training jobs by using the cluster.

D. Create an AWS Batch job that uses Amazon EC2 Spot Instances. Queue the training jobs by using FIFO queues.

Correct Answer: B

Explanation:

This solution uses SageMaker AI managed warm pools. You can use SageMaker AI managed warm pools for quick sequential runs of the training jobs. This solution keeps the cluster warm between training jobs. This solution is serverless and therefore minimizes infrastructure costs. You pay only for the time the training job is active or the warm pool is waiting for the next job. Therefore, this solution is suitable for use cases with intermittent workloads. This solution starts each job as soon as the previous one finishes and is cost-effective on days with no training jobs.

Question 33

An ML engineer is building a classification model. During data exploration, the ML engineer notices that one of the columns is right-skewed.

- A. Quantile binning
- B. Cartesian product transformation
- ✓ C. **Logarithm transformation**
- D. Orthogonal sparse bigram (OSB) transformation

Correct Answer: C

Explanation:

Logarithm transformation is an effective technique to mitigate right-skewed data. You can apply a logarithmic function to the skewed column. Then, the transformation can help normalize the distribution. This solution will make the column more symmetric and suitable for many ML algorithms. This solution is a common approach in data preprocessing to handle skewed numerical data.

Question 34

A company wants to deploy a new ML model to Amazon SageMaker AI. The company developed the model by using the JAX framework. The JAX framework must be immediately available in a SageMaker AI notebook to all notebook users.

- A. Store the JAX framework and associated libraries in AWS CodeArtifact. Install the framework and libraries by using the SageMaker SDK.
- B. Use the prebuilt PyTorch SageMaker AI Docker container with a requirements.txt file to download and install the JAX framework and associated libraries.
- C. Use a PyTorch SageMaker AI notebook. Install the JAX framework and associated libraries before using the model. Ensure that the correct runtime is available.

✓ **D. Create a custom Docker container that includes the JAX framework and associated libraries. Push the container image to Amazon Elastic Container Registry (Amazon ECR). Use the container image in SageMaker AI.**

Correct Answer: D

Explanation:

Amazon ECR is a service that you can use to store Docker images that use repositories. You should create a custom Docker container with the JAX framework and associated libraries to add a custom framework to SageMaker AI. This solution meets the requirement to make JAX immediately available to all notebook users. You can push the container image to Amazon ECR and use the container image in SageMaker AI. This solution includes all necessary dependencies and ensures that the environment is consistent for all users.

Question 35

A company deployed into production an ML model to use for image classification. The company developed a new version of the model with improved precision. The company wants to deploy the new model into production. The company must test the new model alongside the old model by using the same data. Precision cannot be compromised. Some downtime can be tolerated.

- A. Blue/green deployment
- B. Canary deployment
- C. A/B testing

✓ **D. Shadow testing**

Correct Answer: D

Explanation:

Shadow testing makes the new version of the model available alongside the old version. The input data runs through both versions. This strategy performs inference on the same data. This strategy ensures that the new model works as expected while minimizing risk.

Question 36

An industrial company is exploring the use of generative AI to retrieve information from maintenance notes. The company currently uses a text-to-text foundation model (FM) on Amazon Bedrock. The company wants to optimize the model. The company wants to help the model better understand the terminology and abbreviations that maintenance technicians use.

- A. Replace the model with a new model that has more parameters.
- B. Create an Amazon Bedrock Knowledge Base.
- C. Pre-train a new FM by using the Amazon SageMaker AI model parallelism library.

✓ **D. Customize the model by using fine-tuning.**

Correct Answer: D

Explanation:

Fine-tuning is the process of customizing a model to improve its performance in a particular task. You can use fine-tuning to improve a model's domain-specific knowledge within a particular industry by providing labeled examples of industry-specific terminology. This solution is the most cost-effective because the process will take less time than continued pre-training.

Question 37

A company is working on an ML project that will include Amazon SageMaker AI notebook instances. An ML engineer must ensure that the SageMaker AI notebook instances do not allow root access.

✓ **A. Use IAM condition keys to stop deployments of SageMaker AI notebook instances that allow root access.**

B. Monitor resource creation by using Amazon EventBridge events. Create an AWS Lambda function that deletes all deployed SageMaker AI notebook instances that allow root access.

C. Monitor resource creation by using AWS CloudFormation events. Create an AWS Lambda function that deletes all deployed SageMaker AI notebook instances that allow root access.

D. Use AWS Key Management Service (AWS KMS) keys to stop deployments of SageMaker AI notebook instances that allow root access.

Correct Answer: A

Explanation:

IAM condition keys are attributes that you can include in an IAM policy statement to define conditions to grant or deny permissions. You can use IAM condition keys to prevent the creation of SageMaker AI notebook instances that do not have the required security controls. Specifically, you can use the condition key `sagemaker:RootAccess` in an IAM policy to deny the creation of notebook instances with root access.

Question 38

A company uses Amazon SageMaker AI for its ML process. A compliance audit discovers that an Amazon S3 bucket for training data uses server-side encryption with S3 managed keys (SSE-S3). The company requires customer managed keys. An ML engineer changes the S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). The ML engineer makes no other configuration changes. After the change to the encryption settings, SageMaker AI training jobs start to fail with AccessDenied errors.

- A. Update the IAM policy that is attached to the execution role for the training jobs. Include the s3:ListBucket and s3:GetObject permissions.
- B. Update the IAM policy that is attached to the user that created the training jobs. Include the kms:CreateGrant permission.
- ✓ **C. Update the IAM policy that is attached to the execution role for the training jobs. Include the kms:Encrypt and kms:Decrypt permissions.**
- D. Update the S3 bucket policy that is attached to the S3 bucket. Set the value of the aws:SecureTransport condition key to True.

Correct Answer: C

Explanation:

When you use SSE-KMS, the IAM execution role for SageMaker AI training jobs needs kms:Encrypt and kms:Decrypt permissions to access the AWS KMS encrypted S3 data. Adding these permissions will allow SageMaker AI to read and write the encrypted training data.

Question 39

A company wants to improve the performance of its ML model that is deployed in production. The company evaluates the performance of the model by getting feedback from customers. When the company deploys a new version of the model, only a small subset of customers should use the new version. After the company collects feedback and validates the model, the company needs to make the model available to more customers.

A. Use Amazon SageMaker AI endpoints. Create two production model variants. Assign a larger weight to the new model than to the existing model. Decrease the new model's weight after validation occurs.

B. Use an Amazon SageMaker AI endpoint. Create a production variant and a shadow variant. Configure the new model as the shadow variant. Assign a larger traffic sampling percentage to the new model than to the existing model. Decrease the new model's traffic sampling percentage after validation occurs.

✓ **C. Use Amazon SageMaker AI endpoints. Create two production model variants. Assign a smaller weight to the new model than to the existing model. Increase the new model's weight after validation occurs.**

D. Use an Amazon SageMaker AI endpoint. Create a production variant and a shadow variant. Configure the new model as the shadow variant. Assign a smaller traffic sampling percentage to the new model than to the existing model. Increase the new model's traffic sampling percentage after validation occurs.

Correct Answer: C

Explanation:

You can use a SageMaker AI endpoint with two production model variants for A/B testing of ML models. By assigning a smaller weight to the new model, only a subset of customers will use the new model initially. This solution meets the requirement to expose the new model to a small group first. You can increase the weight after validation to make the new model available to more customers. This solution provides a direct comparison of model performance and the collection of customer feedback.

Question 40

A data scientist is building a translation model by using Amazon SageMaker AI seq2seq. The model will translate news articles into multiple languages. The data scientist has a dataset with historical news articles written in multiple languages. The data scientist developed two different models and wants to compare the effectiveness of the models.

- A. F1 score
- B. Recall
- ✓ C. Bilingual evaluation understudy (BLEU) score
- D. Root Mean Square Error (RMSE)

Correct Answer: C

Explanation:

You can use BLEU scores to evaluate machine translation models. You can use this metric to assess the quality of translation by comparing the output of the model to the reference translations.

Question 41

A company is using Amazon SageMaker AI to develop ML models. All of the company's resources are in the us-east-1 AWS Region. The company stores sensitive training data in an Amazon S3 bucket. The model training must have network isolation from the internet.

✓ **A. Run the SageMaker AI training jobs in private subnets. Create an S3 interface endpoint. Use the VPC endpoint to access the training data.**

B. Encrypt traffic to Amazon S3 by using a bucket policy that includes a value of True for the `aws:SecureTransport` condition key. Use default at-rest encryption for Amazon S3. Encrypt SageMaker AI instance storage by using server-side encryption with AWS KMS keys (SSE-KMS).

C. Run the SageMaker AI training jobs in public subnets that have an attached security group. In the security group, use inbound rules to limit traffic from the internet. Encrypt SageMaker AI instance storage by using server-side encryption with AWS KMS keys (SSE-KMS).

D. Run the SageMaker AI training jobs in private subnets. Create a NAT gateway. Route traffic for training through the NAT gateway.

Correct Answer: A

Explanation:

You can run SageMaker AI training jobs in private subnets and use an S3 interface endpoint. This solution ensures that all traffic between SageMaker AI and the S3 bucket remains within the AWS network. This solution avoids traversing the public internet. This solution provides network isolation from the internet during model training and allows access to the required S3 data.

Question 42

An ML engineer trained an ML model by using a tabular dataset. The dataset contains 1,000 data points with 5,000 features. The ML engineer needs to reduce overfitting to improve the model's performance.

✓ **A. Reduce the number of features by applying principal component analysis (PCA).**

B. Reduce the number of data points by randomly dropping 100 data points.

C. Increase the number of features by deriving new features from the existing features.

D. Increase the number of data points by collecting 100 additional data points.

Correct Answer: A

Explanation:

PCA is a dimensionality reduction technique that calculates a new set of features. PCA can reduce the number of features while retaining as much variability in the data as possible. By transforming the original features into principal components, PCA minimizes the feature space. Therefore, PCA addresses the root cause of overfitting. PCA is effective in use cases where the dataset contains many more features than data points. Having many more features than data points increases the likelihood of overfitting because of data sparsity.

Question 43

A company has an ML model that needs to run one time each night to predict stock values. The model input is 3MB of data that is collected during the current day. The model produces the predictions for the next day. The prediction process takes less than 1 minute to finish.

A. Use a real-time endpoint. Configure an auto scaling policy to scale the model to 0 when the model is not in use.

✓ **B. Use a serverless inference endpoint. Set the MaxConcurrency parameter to 1.**

C. Use a multi-model serverless endpoint. Enable caching.

D. Use an asynchronous inference endpoint. Set the InitialInstanceCount parameter to 0.

Correct Answer: B

Explanation:

You can use serverless inference to deploy and scale ML models without the need to configure or manage the underlying infrastructure. With serverless inference, you pay for only the compute time that you use during inference. Serverless inference automatically scales to zero when not in use. The MaxConcurrency parameter specifies the maximum number of concurrent requests that the endpoint can process. Setting the MaxConcurrency parameter to 1 would make the endpoint process one request at a time. The model is invoked only one time each day and takes less than 1 minute to run. Therefore, a MaxConcurrency parameter of 1 is suitable.

Question 44

A company's data scientists are using Amazon SageMaker Studio to train and implement new ML models. The company needs to allocate costs for each user. The company decides to create a report in AWS Cost Explorer that details daily costs with a tag filter.

- A. Create an AWS Lambda function that adds a new createdBy tag for each resource in SageMaker AI. Wait 24-48 hours for tags to show up in the cost allocation tag panel.
- B. Create an AWS Lambda function that adds a new createdBy tag for each resource in SageMaker AI. Wait 1 month for tags to show up in the cost allocation tag panel.

✓ **C. Activate the following cost allocation tag: sagemaker:<user-profile-arn>. Wait 24-48 hours for tags to show up in the cost allocation tag panel.**

- D. Activate the following cost allocation tag: sagemaker:<domain-arn>. Wait 1 month for tags to show up in the cost allocation tag panel.

Correct Answer: C

Explanation:

SageMaker Studio has a built-in cost allocation feature that automatically tags all resources with the sagemaker: tag after you activate the feature. The tag identifies the individual user who created each resource. After activation, the tag will be applied automatically to new resources. Therefore, this solution provides detailed cost tracking with no additional operational overhead. Within 24-48 hours of activation, the company can begin generating user-level cost reports in Cost Explorer. Therefore, this solution meets the requirement to allocate costs for each user with minimal operational overhead.

Question 45

A company uses Amazon SageMaker AI to develop an ML model. The company needs to receive a notification if a significant drift in model quality occurs.

- A. Use SageMaker Model Monitor to collect model metrics.
- B. Create a step in SageMaker Data Wrangler to send a notification when the model metrics exceed a threshold.
- C. Collect Amazon CloudWatch logs in an Amazon S3 bucket. Monitor the model metrics.
- D. Use SageMaker Data Wrangler to periodically check model metrics for feature drift.
- E. Create an Amazon CloudWatch alarm that notifies an Amazon Simple Notification Service (Amazon SNS) topic when the model metrics exceed a threshold.

Correct Answer: A, E

Explanation:

SageMaker Model Monitor is a tool that you can use to monitor the quality of ML models in production. You can use SageMaker Model Monitor to set up continuous monitoring for a real-time endpoint. Additionally, you can run on-schedule monitoring for irregularly run batch transform jobs. You can use CloudWatch metrics to create an alarm when a specific metric exceeds a threshold that you specify. For every state that a CloudWatch alarm can take, you can configure the alarm to send a message to an SNS topic.

Question 46

A company has deployed an image classification ML model. The company improves the existing model's accuracy. An ML engineer needs to validate the model upgrade in a production environment. The ML engineer must not change the existing production endpoint and must not expose end users to the upgraded model.

- A. Deploy the upgraded model as a production variant. Split the production model and the production variant between two endpoints.
- B. Deploy the upgraded model as a shadow variant. Split the production model and the shadow variant between two endpoints.
- C. Deploy the upgraded model as a production variant. Join the production model and the production variant behind a unique endpoint.
- ✓ **D. Deploy the upgraded model as a shadow variant. Join the production model and the shadow variant behind a unique endpoint.**

Correct Answer: D

Explanation:

You can use a shadow variant to test a new model version without exposing end users to the responses. This solution does not change the existing production endpoint. You can use this solution to validate the model upgrade in the production environment without impacting end users. Shadow variants receive a copy of the incoming requests but do not return responses to the users.

Question 47

An ML engineer is using the Amazon SageMaker AI XGBoost algorithm to develop a diagnostic tool. The tool uses multiple indicators to predict whether a patient has a specific disease. The dataset includes data from a large number of patients classified as healthy and a small number of patients diagnosed with the disease. Initial tests show that the model can consistently identify a patient who does not have the disease. However, the model cannot accurately predict whether a patient has the disease.

A. Increase the learning rate.

✓ **B. Increase the value of the `scale_pos_weight` hyperparameter.**

C. Decrease the value of the `scale_pos_weight` hyperparameter.

D. Decrease the value of the `max_depth` hyperparameter.

Correct Answer: B

Explanation:

You can have an imbalanced dataset where one class is underrepresented compared to the other class. In this scenario, the class of patients with the disease is underrepresented compared to the class of healthy patients. You can increase the `scale_pos_weight` hyperparameter to help the model better identify the minority class. Increasing this hyperparameter will give higher weight to the positive class (patients with the disease) during training. This adjustment will effectively increase the penalty for misclassifying instances of the minority class.

Question 48

A company wants to migrate an ML application to the AWS Cloud. The application predicts employee turnover. The company wants to continue using its current code implementation that is based on Apache Spark.

- A. Run the code in an Amazon EMR cluster.
- B. Run the code in Amazon SageMaker Data Wrangler.
- C. Run the code in an Amazon SageMaker Spark Processing job.

✓ **D. Run the code in an AWS Glue interactive session.**

Correct Answer: D

Explanation:

AWS Glue interactive sessions provide an on-demand, serverless Spark runtime environment. Because AWS Glue is serverless, you do not need to manage the underlying infrastructure. Therefore, this solution requires the least operational overhead.

Question 49

An ML engineer needs to develop an audio podcast from written scripts. The ML engineer must introduce a unique break after each paragraph of the script.

A. Use Amazon Transcribe to convert the text to speech. Define the breaks by using SSML.

✓ **B. Use Amazon Polly to convert the text to speech. Define the breaks by using Speech Synthesis Markup Language (SSML).**

C. Use Amazon Comprehend to convert the text to speech. Define the breaks by using Speech Synthesis Markup Language (SSML).

D. Use Amazon Polly to convert the text to speech. Define the breaks by using lexicons.

Correct Answer: B

Explanation:

Amazon Polly is a service that you can use to convert text to speech. You can use Amazon Polly to develop an audio podcast from written scripts. SSML in Amazon Polly provides precise control over speech output, including the ability to introduce breaks. By using SSML tags, you can define unique breaks after each paragraph of the script.

Question 50

A company has a trained ML model. The company needs an inference option that is always available. The inference option also must provide low, consistent latency for predictable traffic patterns.

- A. Serverless inference
- B. Batch inference
- ✓ **C. Real-time inference**
- D. Asynchronous inference

Correct Answer: C

Explanation:

Real-time inference is designed to provide low and consistent latency for predictable traffic patterns. Real-time inference ensures that the model is always available. Real-time inference is suitable for applications that require immediate responses.

Question 51

A company created an ML training job by using the Amazon SageMaker AI API. The company stores training data on Amazon S3. The company wants to replicate a subset of the data onto each ML compute instance that is launched for model training.

- A. Set the S3DataType field to FullyReplicated.
- B. Set the S3DataDistributionType field to FullyReplicated.
- ✓ **C. Set the S3DataDistributionType field to ShardedByS3Key.**
- D. Set the S3Uri field to FullyReplicated.

Correct Answer: C

Explanation:

You can use the S3DataDistributionType field to designate how the dataset is replicated on each ML compute instance. To replicate a subset of data on each ML compute instance that is launched for model training, you can set the S3DataDistributionType field to ShardedByS3Key.

Question 52

A data scientist wants to use generative AI to run natural language queries on thousands of legal documents and generate responses.

✓ **A. Use the Amazon Kendra Retrieve API to search an Amazon Kendra index to return relevant text chunks. Pass relevant text chunks to a language model for generative responses.**

B. Use the Amazon Lex console to create an Amazon Lex bot to query the documents to return both relevant text chunks and generative responses.

C. Use Amazon Comprehend entity recognition to identify relevant text chunks. Pass relevant text chunks to a language model for generative responses.

D. Use the Amazon Bedrock API to customize a foundation model (FM) for generative responses. Query the documents to return relevant text chunks.

Correct Answer: A

Explanation:

Amazon Kendra is a service that uses natural language processing (NLP) and semantic and contextual understanding for intelligent search. You can use the Retrieve API within Amazon Kendra to first retrieve the most relevant text chunks across documents. Then, the API will pass the text chunks to the language model to provide the best responses.

Question 53

A company needs to host a custom ML model to perform forecast analysis. The forecast analysis will occur with predictable and sustained load during the same 2-hour period every day. Multiple invocations during the analysis period will require quick responses. The company needs AWS to manage the underlying infrastructure and any auto scaling requirements.

- A. Schedule an Amazon SageMaker batch transform job by using AWS Lambda.
- B. Configure an Auto Scaling group of Amazon EC2 instances to use scheduled scaling.
- ✓ **C. Use Amazon SageMaker Serverless Inference with provisioned concurrency.**
- D. Run the model on an Amazon Elastic Kubernetes Service (Amazon EKS) cluster on Amazon EC2 with pod auto scaling.

Correct Answer: C

Explanation:

Serverless inference is an inference option that automatically provisions, scales, and manages compute capacity to run your ML model. You can configure serverless inference with provisioned concurrency. This solution provides low-latency responses for predictable workloads by keeping the model warm and ready to serve requests.

Question 54

A company is planning to create several ML prediction models. The training data is stored in Amazon S3. The entire dataset is more than 5TB in size and consists of CSV, JSON, Apache Parquet, and simple text files. The data must be processed in several consecutive steps. The steps include complex manipulations that can take hours to finish running. Some of the processing involves natural language processing (NLP) transformations. The entire process must be automated.

- A. Process data at each step by using Amazon SageMaker Data Wrangler. Automate the process by using SageMaker Data Wrangler jobs.
- B. Use Amazon SageMaker AI notebooks for each data processing step. Automate the process by using Amazon EventBridge.
- C. Process data at each step by using AWS Lambda functions. Automate the process by using AWS Step Functions and Amazon EventBridge.

✓ **D. Use Amazon SageMaker Pipelines to create a pipeline of data processing steps. Automate the pipeline by using Amazon EventBridge.**

Correct Answer: D

Explanation:

SageMaker Pipelines is a tool that you can use to create and manage ML workflows at scale. You can use SageMaker Pipelines to process large datasets with complex, multi-step transformations. SageMaker Pipelines can handle more than 5TB of diverse data formats that are stored in Amazon S3. The data formats include CSV, JSON, Parquet, and text files. You can use SageMaker Pipelines to create a series of interconnected processing steps for complex manipulations and NLP transformations. You can automate the pipeline by using EventBridge. This solution provides the scalability, flexibility, and automation that you need to create multiple ML prediction models.

Question 55

An ML engineer receives datasets that contain missing values, duplicates, and extreme outliers. The ML engineer must consolidate these datasets into a single data frame and must prepare the data for training.

- A. Use Amazon SageMaker Ground Truth to import the datasets and to consolidate them into a single data frame. Use the human-in-the-loop capability to prepare the data.
- B. Manually import and merge the datasets. Consolidate the datasets into a single data frame. Use Amazon SageMaker AI data labeling to prepare the data.
- ✓ **C. Use Amazon SageMaker Data Wrangler to import the datasets and to consolidate them into a single data frame. Use the cleansing and enrichment functionalities to prepare the data.**
- D. Manually import and merge the datasets. Consolidate the datasets into a single data frame. Use Amazon Q Developer to generate code snippets that will prepare the data.

Correct Answer: C

Explanation:

You can use SageMaker Data Wrangler to prepare data for ML workflows. You can use SageMaker Data Wrangler to import and consolidate the datasets into a single frame. Then, you can cleanse, analyze, and transform the data for training.

Question 56

A company runs Amazon SageMaker AI ML models that use accelerated instances. The models require real-time responses. Each model has different scaling requirements. The company must not allow a cold start for the models.

✓ **A. Create a SageMaker AI endpoint. Create an inference component for each model. In the inference component settings, specify the newly created endpoint. Create an auto scaling policy for each inference component. Set the parameter for the minimum number of copies to at least 1.**

B. Create an Amazon S3 bucket. Store all the model artifacts in the S3 bucket. Create a SageMaker AI multi-model endpoint. Point the endpoint to the S3 bucket. Create an auto scaling policy for the endpoint. Set the parameter for the minimum number of copies to at least 1.

C. Create a SageMaker Asynchronous Inference endpoint for each model. Create an auto scaling policy for each endpoint.

D. Create a SageMaker Serverless Inference endpoint for each model. Use provisioned concurrency for the endpoints.

Correct Answer: A

Explanation:

SageMaker AI inference components are a feature that you can use to configure and manage individual models within a SageMaker AI endpoint. You can use accelerated instances when you create a SageMaker AI endpoint with inference components for each model. The auto scaling policy for each inference component accommodates different scaling requirements for each model. You can set the minimum number of copies to at least 1 to ensure that at least 1 instance is always running. This solution avoids cold starts and meets the real-time response requirement.

Question 57

A company needs to use retrieval augmented generation (RAG) to supplement an open source large language model (LLM) that runs on Amazon Bedrock. The company's data for RAG is a set of documents in an Amazon S3 bucket. The documents consist of .csv files and .docx files.

- A. Fine-tune an existing LLM by using an AutoML job in Amazon SageMaker AI. Configure the S3 bucket as a data source for the AutoML job. Deploy the LLM to a SageMaker AI endpoint. Use the endpoint to perform RAG queries.
- B. Convert the data into vectors. Store the data in an Amazon Neptune database. Connect the database to Amazon Bedrock. Call the Amazon Bedrock API to perform RAG queries.
- C. Create a pipeline in Amazon SageMaker Pipelines to generate a new model. Call the new model from Amazon Bedrock to perform RAG queries.

✓ **D. Create an Amazon Bedrock knowledge base. Configure a data source that references the S3 bucket. Use the Amazon Bedrock API to perform RAG queries.**

Correct Answer: D

Explanation:

You can use Amazon Bedrock knowledge bases to search data to find the most useful information. You can create an Amazon Bedrock knowledge base and configure the knowledge base to use the S3 bucket as a data source. This method provides a fully managed RAG solution. This solution supplements the LLM with information from the .csv and .docx files in Amazon S3 without additional data movement or complex pipeline management. By using the Amazon Bedrock API to perform RAG queries, you can integrate the knowledge base with the existing LLM.

Question 58

A company needs to run a batch data-processing job on Amazon EC2 instances. The job will run during the weekend and will take 90 minutes to finish running. The processing can handle interruptions. The company will run the job every weekend for the next 6 months.

- A. On-Demand Instances
- B. Reserved Instances
- ✓ **C. Spot Instances**
- D. Dedicated Instances

Correct Answer: C

Explanation:

Spot Instances are a purchasing option that provides available EC2 instance capacity at a discounted price. Spot Instances are suitable for workloads that can handle interruptions. The job takes only 90 minutes and can handle interruptions. Therefore, Spot Instances are the most cost-effective option.

Question 59

An ML engineer has deployed a sentiment analysis ML model on an Amazon SageMaker AI endpoint that is supported by Amazon EC2 instances. The model will go into production in 1 week. The ML engineer must implement a solution to accommodate large increases and decreases in user traffic.

- A. Increase the number of EC2 instances to meet the expected peak demand.
- B. Deploy an Application Load Balancer (ALB). Register the endpoint as a target for the ALB.
- ✓ **C. Configure a target tracking scaling policy based on a metric in Amazon CloudWatch.**
- D. Configure the EC2 instances to have more GPU compute capabilities.

Correct Answer: C

Explanation:

A target tracking scaling policy can automatically adjust the number of instances that are provisioned based on inference requests and workload. You can configure a target tracking scaling policy that is based on a metric in CloudWatch. Then, the policy can accommodate large increases and decreases in user traffic for the SageMaker AI endpoint. This solution provides dynamic scaling of EC2 instances based on real-time metrics. Therefore, this solution can ensure that the number of instances support the actual demand.

Question 60

A company wants to encrypt all data in transit when an ML training job runs. The ML engineer must ensure that encryption in transit is applied to processes that Amazon SageMaker AI uses during the training job.

✓ **A. Encrypt communication between nodes in a training cluster.**

B. Specify an AWS Key Management Service (AWS KMS) key during the creation of the SageMaker AI domain.

C. Specify an AWS Key Management Service (AWS KMS) key during the creation of the training job request.

D. Encrypt communication between nodes for batch processing by using batch transform.

Correct Answer: A

Explanation:

SageMaker AI can encrypt inter-node communication when you run distributed training jobs. This solution ensures the encryption of all the data that is transmitted between nodes in a training cluster. Therefore, this solution meets the requirement to encrypt data in transit during training.

Question 61

A glass manufacturing company uses a computer vision model to detect broken glass on its continuous production lines. The company uses Amazon SageMaker AI to train the model with more than 50GB of images that are stored in Amazon S3. The company has deployed the model to the cloud. The model is re-trained daily, but the training job starts more slowly than the company expects. The company needs to reduce the startup time of the training job.

✓ **A. Use pipe mode as the input mode for the training job configuration.**

B. Load the training data onto an Amazon Elastic Block Store (Amazon EBS) volume. Attach the EBS volume to the training job instance.

C. Change the instance type of the Amazon EC2 instance that the training job uses.

D. Use an Amazon FSx for Lustre file system to stream data from Amazon S3 to the training job.

Correct Answer: A

Explanation:

Pipe mode is an input mode for SageMaker AI training jobs that streams data directly from Amazon S3. This streaming approach can significantly reduce startup times for training jobs. This approach is especially helpful for large datasets, like the 50GB of images in the scenario. Pipe mode is the most cost-effective option to meet the requirement to reduce training job startup time. Pipe mode does not require any additional infrastructure or services. Therefore, this approach is the most suitable for daily retraining.

Question 62

A company wants to use Amazon Bedrock to automatically generate customer-facing content. The content must concisely explain the company's operational rules, special events, and guidelines. All content that is generated by Amazon Bedrock must be consistent with the company's brand guidelines. The company plans to use the Claude family of models by Anthropic on Amazon Bedrock.

- A. Use higher top_p values.
- B. Use higher top_k values.
- C. Use specific stop_sequences (string).
- D. Use higher temperature values.
- E. Control length by using max_tokens_to_sample (int).

Correct Answer: C, E

Explanation:

The stop_sequences parameter provides the precise marking of content endpoints. By using clear endpoints, the generated content remains organized and consistent. This parameter can match the company's brand guidelines. The max_tokens_to_sample parameter provides direct control over the length of the generated text. You can use this parameter to ensure that the output is brief and ends at an appropriate length. This parameter is useful for creating concise messages that need to match information like operational rules, special events, and brand guidelines.

Question 63

A company needs a model that predicts the price of a house before the house is listed for sale.

A. Principal component analysis (PCA)

✓ **B. XGBoost for regression**

C. Semantic segmentation

D. Instance segmentation

Correct Answer: B

Explanation:

The XGBoost algorithm is an open source implementation of the gradient boosting technique. The gradient boosting technique is a supervised learning method that combines predictions from multiple simpler models to accurately forecast a target variable. Predicting house prices is a regression problem. You can use XGBoost to handle complex relationships between various features. Features can include location, amenities, and the target variable (house price). XGBoost for regression can predict the price of a house before the house is listed for sale by learning from historical data and generating accurate numerical predictions.

Question 64

An ecommerce company designed an ML recommendation system to suggest similar products to customers that visit a product page. The system was deployed by using an Amazon SageMaker AI endpoint. After the system deployment, traffic has increased sporadically throughout the day. An ML engineer must implement a solution that uses auto scaling policies to better handle the increasing traffic without compromising customer experience.

✓ **A. Use a target tracking scaling policy based on a custom metric and a target value to scale instance count.**

B. Use a target tracking scaling policy to specify different instance types based on a predefined metric.

C. Use a schedule-based scaling policy to scale instance count at different times during the day.

D. Use a step scaling policy to specify different instance types based on predefined metrics.

Correct Answer: A

Explanation:

A target tracking scaling policy targets a metric and automatically adds or removes instances based on the specified target value. You can use predefined metrics or custom metrics as a target. This solution ensures the system can handle the increasing traffic without compromising customer experience.

Question 65

An ML engineer needs to optimize features to train ML models. Select the correct feature engineering technique from the following list for each use case. Each feature engineering technique should be selected one time or not at all. (Note: The matching answer table was interactive and was not captured in the saved page.)

(Matching question - interactive table not captured in the report)

Correct Answer: N/A

Explanation:

Matching question - answer table was not captured in the Skill Builder report.